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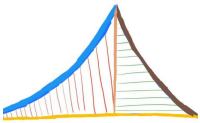

G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

Contents

1 Signal Processing 4
1.1 Analog 4
1.2 Discrete 9

2 Optimization 12

3 Probability 12

1 Signal Processing

1.1 Analog

1.1.1 A spring-mass-damper system with a mass of 1 kg is found to have a damping ratio of 0.2 and a natural frequency of 5 rad/s. The damping of the system is given by (AE 2007)

a) 2 Ns/m b) 2 N/s c) 0.2 kg/s d) 0.2 N/m

1.1.2 A 1 kg mass attached to a spring elongates it by 16 mm. The mass is then pulled from its equilibrium position by 10 mm and released from rest. Assuming the acceleration due to gravity of $9.81 m/s^2$, the response of the mass in mm is given by: (AE 2007)

a) $x = \cos 4.76\,t$ b) $x = \sin 4.76\,t$ c) $x = \sin 16\,t$ d) $x = 10 \cos 16\,t$

1.1.3 A spring-mass-damper system is excited by a force $F_0 \sin \omega t$. The amplitude at resonance is measured to be 1 cm. At half the resonance frequency, the amplitude is 0.5 cm. The damping ratio of the system is (AE 2007)

a) 0.1026 b) 0.3242 c) 0.7211 d) 0.1936

1.1.4 $\lim_{x \to 0} \frac{\sin x}{e^4 x} =$ (AE 2007)

a) 10 b) 0 c) 1 d) ∞

1.1.5 Let a dynamical system be described by the differential equation $2\frac{dx}{dt} + \cos x = 0$. Which of the following differential equations describes this system in a close approximation sense for small perturbation about $x = \pi/4$? (AE 2007)

a) $2\frac{dx}{dt} + \sin x = 0$
b) $2\frac{dx}{dt} - \frac{1}{\sqrt{2}} x = 0$
c) $\frac{dx}{dt} + \cos x = 0$
d) $\frac{dx}{dt} + x = 0$

1.1.6 Consider the spring mass system shown in the figure below. This system has two degrees of freedom representing the motions of the two masses. (AE 2007)

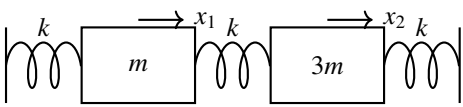

Fig. 1.1.6.1

a) The system shows the following type of coordinate coupling
   i) Static coupling
   ii) Dynamic coupling
   iii) Static and dynamic coupling
   iv) No coupling
b) The two natural frequencies of the system are given as

i) $\sqrt{\frac{4\pm\sqrt{5}k}{3m}}$ ii) $\sqrt{\frac{4\pm\sqrt{3}k}{3m}}$ iii) $\sqrt{\frac{4\pm\sqrt{7}k}{3m}}$ iv) $\sqrt{\frac{4\pm\sqrt{11}k}{3m}}$

1.1.7 Let

$$F(s) = \frac{(s+10)}{(s+2)(s+20)}$$

(AE 2007)

a) The partial fraction expansion of $F(s)$ is

i) $\frac{1}{s+2} + \frac{1}{s+20}$ ii) $\frac{5}{s+2} + \frac{2}{s+20}$ iii) $\frac{2}{s+2} + \frac{20}{s+20}$ iv) $\frac{4/9}{s+2} + \frac{5/9}{s+20}$

b) The inverse Laplace transform of $F(s)$ is

i) $2e^{-34} + 20e^{-34}$
ii) $\frac{4}{9}e^{-34} + \frac{5}{9}e^{-34}$
iii) $5e^{-34} + 2e^{-34}$
iv) $\frac{9}{4}e^{-34} + \frac{9}{5}e^{-34}$

1.1.8

$$\frac{\omega}{(s+a)^2+\omega^2}$$

is the Laplace Transform of (AG 2007)

a) $\exp(-at)\sin\omega t$
b) $\sin\omega t$
c) $\exp(-at)\sinh\omega t$
d) $\sinh\omega t$

1.1.9 Evaluate $\int_0^{\pi} \frac{\sin t}{t} dt$ (CE 2007)

a) π b) $\pi/2$ c) $\pi/4$ d) $\pi/8$

1.1.10 The RC circuit shown is:

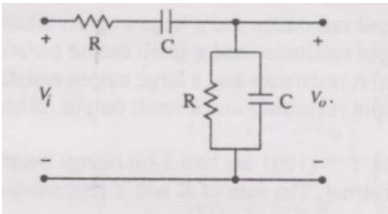

Fig. 1.1.10.1

(EC 2007)

a) Low-pass filter
b) High-pass filter
c) Band-pass filter
d) Band-reject filter

1.1.11 The correct full wave rectifier circuit is:

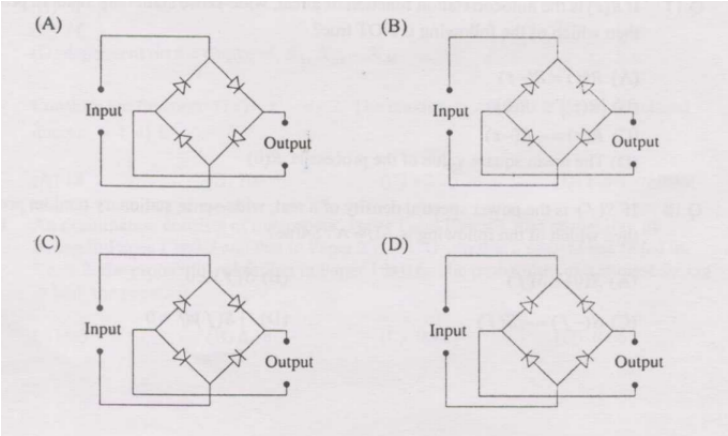

Fig. 1.1.11.1

1.1.12 Given closed-loop transfer function $T(s) = \frac{s-5}{(s+2)(s+3)}$, the system is: (EC 2007)

a) Unstable
b) Uncontrollable
c) Minimum phase
d) Non-minimum phase

1.1.13 If Laplace transform of $y(t)$ is $Y(s) = \frac{1}{s(s-1)}$, the final value is: (EC 2007)

a) -1 b) 0 c) 1 d) Unbounded

1.1.14 The solution of the differential equation $k^2 \frac{d^2y}{dx^2} = y$, under the boundary conditions (i) $y = y_1$ at $x = 0$ and (ii) $y = y_2$ at $x = \infty$, where k, y_1, y_2 are constants, is: (EC 2007)

(A) $y = (y_1 - y_2)\exp(-x/k) + y_2$
(B) $y = (y_1 - y_2)\exp(-x/k) + y_1$
(C) $y = (y_1 - y_2)\sinh(x/k) + y_2$
(D) $y = (y_1 - y_2)\exp(-x/k) + y_1$

1.1.15 Two series resonant filters are as shown in the figure. Let the 3-dB bandwidth of Filter 1 be B_1 and that of Filter 2 be B_2. The value of $\frac{B_1}{B_2}$ is: (EC 2007)

(A) 4 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

1.1.16 In the circuit shown, V_C is 0 volts at $t = 0$ sec. For $t > 0$, the capacitor current $i_c(t)$, where t is in seconds, is given by: (EC 2007)

a) $0.50\exp(-25t)$ mA
b) $0.25\exp(-25t)$ mA

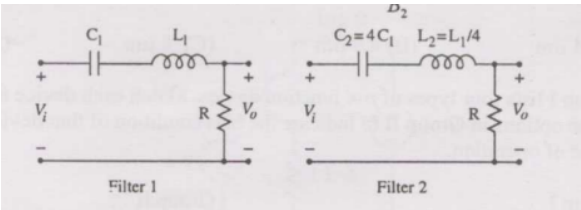

Fig. 1.1.15.1

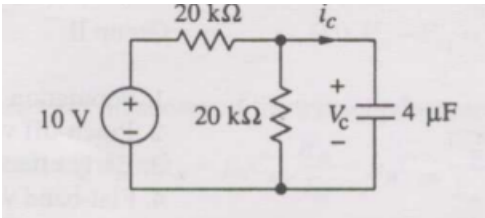

Fig. 1.1.16.1

c) $0.50\exp(-12.5t)$ mA
d) $0.25\exp(-6.25t)$ mA

1.1.17 The 3-dB bandwidth of the low-pass signal $e^{-t}u(t)$ is: (EC 2007)

a) $\frac{1}{2\pi}$ Hz b) $\frac{1}{2\pi}\sqrt{2}-1$ Hz c) ∞ d) 1 Hz

1.1.18 A Hilbert transformer is a: (EC 2007)

a) Non-linear system
b) Non-causal system
c) Time-varying system
d) Low-pass system

1.1.19 The frequency response of a linear time-invariant system is given by $H(f) = \frac{5}{1+j10\pi f}$. The step response of the system is: (EC 2007)

a) $5\left(1-e^{-5t}\right)u(t)$
b) $5\left(1-e^{-t/5}\right)u(t)$
c) $\frac{1}{5}\left(1-e^{-5t}\right)u(t)$
d) $\frac{1}{5}\left(1-e^{-t/5}\right)u(t)$

1.1.20 A control system with a PD controller is shown. If the velocity error constant $K_v = 1000$ and the damping ratio $\zeta = 0.5$, then the values of K_P and K_D are: (EC 2007)

a) $K_P = 100,\ K_D = 0.09$
b) $K_P = 100,\ K_D = 0.9$
c) $K_P = 10,\ K_D = 0.09$
d) $K_P = 10,\ K_D = 0.9$

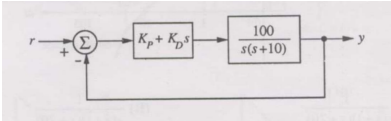

Fig. 1.1.20.1

1.1.21 The transfer function of a plant is $T(s) = \frac{5}{(s+5)(s^2+s+1)}$. The second-order approximation of $T(s)$ using the dominant pole concept is: (EC 2007)

(A) $\frac{1}{(s+5)(s+1)}$ (B) $\frac{5}{(s+5)(s+1)}$ (C) $\frac{5}{s^2+s+1}$ (D) $\frac{1}{s^2+s+1}$

1.1.22 The open-loop transfer function of a plant is given as $G(s) = \frac{1}{s^2-1}$. If the plant is operated in unity feedback, then the lead compensator that can stabilize the system is: (EC 2007)

a) $\frac{10(s-1)}{s+2}$ b) $\frac{10(s+4)}{s+2}$ c) $\frac{10(s+2)}{s+10}$ d) $\frac{2(s+2)}{s+10}$

1.1.23 A unity feedback control system has an open-loop transfer function $G(s) = \frac{K}{s(s^2+7s+12)}$. The gain K for which $s = -1 + j1$ will lie on the root locus of this system is: (EC 2007)

a) 4 b) 5.5 c) 6.5 d) 10

1.1.24 The asymptotic Bode magnitude plot of a transfer function is as shown: starts at 60 dB, slopes −20 dB/decade until 1 rad/s (at 40 dB), then −40 dB/decade crossing 0 dB near 10, then −60 dB/decade. The transfer function $G(s)$ corresponding to this plot is: (EC 2007)

a) $\frac{1}{(s+1)(s+20)}$ b) $\frac{1}{s(s+1)(s+20)}$ c) $\frac{100}{s(s+1)(s+20)}$ d) $\frac{100}{s(s+1)(1+0.05s)}$

1.1.25 The state-space representation of a separately excited DC servo motor dynamics is given as $\frac{d}{dt}\begin{pmatrix}\omega \\ i_a\end{pmatrix} = \begin{pmatrix}-1 & 1 \\ -1 & -10\end{pmatrix}\begin{pmatrix}\omega \\ i_a\end{pmatrix} + \begin{pmatrix}0 \\ 10\end{pmatrix}u$, where ω is the speed, i_a is the armature current, and u is the armature voltage.
The transfer function $\frac{\omega(s)}{u(s)}$ is: (EC 2007)

a) $\frac{10}{s^2+11s+11}$ b) $\frac{1}{s^2+11s+11}$ c) $\frac{10s+10}{s^2+11s+11}$ d) $\frac{1}{s^2+s+1}$

1.1.26 The raised cosine pulse $p(t)$ used for zero ISI with unity roll-off factor is $p(t) = \frac{\sin 4\pi Wt}{4\pi Wt(1-16W^2t^2)}$. The value of $p(t)$ at $t = \frac{1}{4W}$ is: (EC 2007)

a) -0.5 b) 0 c) 0.5 d) ∞

1.1.27 Consider the op-amp circuit shown.

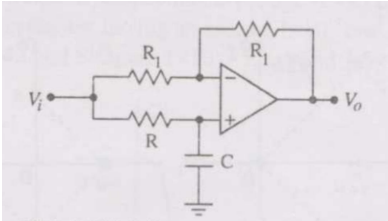

Fig. 1.1.27.1

a) The transfer function $V_o(s)/V_i(s)$ is: (EC 2007)

i) $\dfrac{1 - sRC}{1 + sRC}$ ii) $\dfrac{1 + sRC}{1 - sRC}$ iii) $\dfrac{1}{1 - sRC}$ iv) $\dfrac{1}{1 + sRC}$

b) If $V_i = V_i \sin(\omega t)$ and $V_o = V_o \sin(\omega t + \phi)$, the minimum and maximum values of ϕ (in radians) are respectively: (EC 2007)

i) $-\pi/2$ and $\pi/2$ ii) 0 and $\pi/2$ iii) $-\pi$ and 0 iv) $-\pi/2$ and 0

1.2 Discrete

1.2.1 The Euler iteration formula for numerically integrating a first order nonlinear differential equation of form $x = f(x)$, with a constant step size of Δt is (AE 2007)

a) $x_{k+1} = x_k - \Delta t \times f(x_k)$
b) $x_{k+1} = x_k + \left(\Delta t^2/2\right) \times f(x_k)$
c) $x_{k+1} = x_k - (1/\Delta t) \times f(x_k)$
d) $x_{k+1} = x_k + \Delta t \times f(x_k)$

1.2.2 Numerical values of the integral $\int_0^1 \frac{1}{1+x^2}\, dx$ if evaluated numerically using the Trapazoidal rule with dx = 0.2 would be (AE 2007)

a) 1 b) $\pi/4$ c) 0.7837 d) 0.2536

1.2.3 The Newton-Raphson iteration formula to find a cube root of a positive number c is (AE 2007)

a) $x_{k+1} = \frac{2x_k^3 + \sqrt[3]{c}}{3x_k^2}$ b) $x_{k+1} = \frac{2x_k^3 - \sqrt[3]{c}}{-3x_k^2}$ c) $x_{k+1} = \frac{2x_k^3 + c}{3x_k^2}$ d) $x_{k+1} = \frac{x_k^3 + c}{3x_k^2}$

1.2.4 $y(0) = 0$ and using Euler's method with step size $h = 0.1$ solution of the differential equation

$$\frac{dy}{dx} = 2xy + 1$$

gives the value of $y(0.3) =?$ (AG 2007)

a) 0.3101 b) 0.3142 c) 0.6202 d) 4.080

1.2.5 Integrating the function (AG 2007)

$$f(x) = 1 + e^{-x}\sin(4x)$$

over the interval [0, 1] using Simpson's $\frac{1}{3}$ rule gives Result = ?

a) 1.021 b) 1.091 c) 1.321 d) 2.642

1.2.6 The degree of the differential equation $\frac{d^2x}{dt^2} + 2x^3 = 0$ is (CE 2007)

a) 0 b) 1 c) 2 d) 3

1.2.7 The solution of the differential equation $\frac{dy}{dx} = x^2y$ with the condition that $y = 1$ at $x = 0$ is (CE 2007)

a) $y = e^{\frac{1}{2x}}$ b) $\ln(y) = \frac{x^3}{3} + 4$ c) $\ln(y) = \frac{x^2}{2}$ d) $y = e^{\frac{x^3}{3}}$

1.2.8 The following equation needs to be numerically solved using the Newton-Raphson method. $x^3 + 4x - 9 = 0$ The iterative equation for this purpose is (k indicates the iteration level) (CE 2007)

a) $x_{k+1} = \frac{2x_k^3+9}{3x_k^2+4}$

b) $x_{k+1} = \frac{2x_k^3+9}{3x_k^2+9}$

c) $x_{k+1} = x_k - \frac{x_k^3+4x_k-9}{3x_k^2+4}$

d) $x_{k+1} = \frac{4x_k^2+3}{9x_k^2+2}$

1.2.9 Given that one root of the equation $x^3 - 10x^2 + 31x - 30 = 0$ is 5, the other two roots are (CE 2007)

a) 2 and 3 b) 2 and 4 c) 3 and 4 d) -2 and -3

1.2.10 Consider the series $x_{n+1} = \frac{x_n}{2} + \frac{9}{8x_n}, x_0 = 0.5$ obtained from the Newton-Raphson method. The series converges to (CS 2007)

a) 1.5 b) $\sqrt{2}$ c) 1.6 d) 1.4

1.2.11 The following plot shows a function y which varies linearly with x. The value of the integral $\int_0^2 y\,dx$ is: (EC 2007)

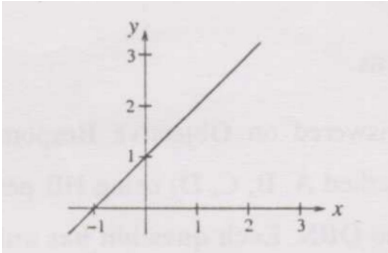

Fig. 1.2.11.1

a) 1.0 b) 2.5 c) 4.0 d) 5.0

1.2.12 For $|x| < 1$, $\coth(x)$ can be approximated as: (EC 2007)

a) x b) x^2 c) $\frac{1}{x}$ d) $\frac{1}{x^2}$

1.2.13 $\lim_{\theta \to 0} \frac{\sin(\theta/2)}{\theta}$ is: (EC 2007)

a) 0.5 b) 1 c) 2 d) Not defined

1.2.14 Which of the following functions is strictly bounded? (EC 2007)

a) $\sin x$ b) e^x c) $\cos x$ d) x^2

1.2.15 For the function e^{-x}, the linear approximation around $x = 2$ is: (EC 2007)

a) $(3 - x)^2$
b) $1 - x$
c) $[3 + 2\sqrt{2} - (1 + \sqrt{2})x]e^2$
d) e^{-2}

1.2.16 The equation $x^2 + x^4 - 4x - 4 = 0$ is to be solved using Newton-Raphson. If $x = 2$ is the initial approximation, then the next approximation using this method will be: (EC 2007)

a) $\frac{2}{3}$ b) $\frac{4}{3}$ c) 1 d) $\frac{3}{2}$

1.2.17 A 5-point sequence $x[n]$ is given as $x[-3] = 1$, $x[-2] = 1$, $x[-1] = 0, x[0] = 5$, $x[1] = 15$. Let $X\left(e^{j\omega}\right)$ denote the discrete-time Fourier transform of $x[n]$. The value of $\int_{-\pi}^{\pi} X\left(e^{4j\omega 3}\right) d\omega$ (EC 2007)

a) 5 b) 10π c) 16π d) $5 + j10\pi$

1.2.18 The z-transform $X[z]$ of a sequence $x[n]$ is given by $X[z] = \frac{0.5}{1-2z^{-1}}$. It is given that the region of convergence of $X[z]$ includes the unit circle. The value of $x[0]$ is: (EC 2007)

a) -0.5 b) 0 c) 0.25 d) 0.5

2 Optimization

2.0.1 The minimum value of

$$J(x) = x^2 - 7x + 30$$

occurs at (AE 2007)

a) $x = 7/2$ b) $x = 7/30$ c) $x = 30/7$ d) $x = 30$

2.0.2 Consider the function $f(x) = x^2 - x - 2$. The maximum value of $f(x)$ in the closed interval $[-4, 4]$ is: (EC 2007)

a) 18 b) 10 c) -2.25 d) Indeterminate

3 Probability

3.0.1 For station X, the maximum one day rainfall with 25 years return period is 100 mm. The probability of a one day rainfall equal to or greater than 100 mm at station X occurring at least once in 15 successive years is (AG 2007)

a) 0.458 b) 0.500 c) 0.637 d) 0.990

3.0.2 If the standard deviation of the spot speed of vehicles in a highway is $8.8kmph$ and the mean speed of the vehicles is $33kmph$, the coefficient of variation in speed is (CE 2007)

a) 0.1517 b) 0.1867 c) 0.2666 d) 0.3646

3.0.3 Suppose we uniformly and randomly select a permutation from the 20! permutations of $1, 2, 3, ..., 20$. What is the probability that 2 appears at an earlier position than any other even number in selected permutation ? (CS 2007)

a) $\frac{1}{2}$ b) $\frac{1}{10}$ c) $\frac{9!}{20!}$ d) None of these

3.0.4 There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot? (CS 2007)

a) $np(1-p)^{n-1}$ b) $(1-p)^{n-1}$ c) $p(1-p)^{n-1}$ d) $1-(1-p)^{n-1}$

3.0.5 If E denotes expectation, the variance of a random variable X is given by: (EC 2007)

a) $E[X^2] - E^2[X]$
b) $E[X^2] + E^2[X]$
c) $E[X^2]$
d) $E^2[X]$

3.0.6 If $R(\tau)$ is autocorrelation of real, WSS random process, which is NOT true: (EC 2007)

a) $R(\tau) = R(-\tau)$
b) $|R(t)| \leq R(0)$
c) $R(\tau) = -R(-\tau)$
d) Mean square value is $R(0)$

3.0.7 If $S(f)$ is power spectral density of a real WSS random process, which is ALWAYS true:

(EC 2007)

a) $S(0) \geq S(f)$
b) $S(f) \geq 0$
c) $S(-f) = -S(f)$
d) $\int S(f)df = 0$

3.0.8 An examination consists of two papers, Paper I and Paper II. The probability of failing in Paper I is 0.3 and that in Paper II is 0.2. Given that a student has failed in Paper II, the probability of failing in Paper I is 0.6. The probability of a student failing in both the papers is: (EC 2007)

a) 0.5
b) 0.18
c) 0.12
d) 0.06

3.0.9 During transmission over a binary channel, bit errors occur independently with probability p. The probability of at most one bit in error in a block of n bits is: (EC 2007)

a) p^n
b) $1 - p^n$
c) $np(1-p)^{n-1} + (1-p)^n$
d) $1 - (1-p)^n$

3.0.10 Two 4-ary signal constellations are shown. It is given that ϕ_1 and ϕ_2 constitute an orthonormal basis and four symbols in each are equiprobable. Let $N_0/2$ denote the power spectral density of white Gaussian noise. (EC 2007)

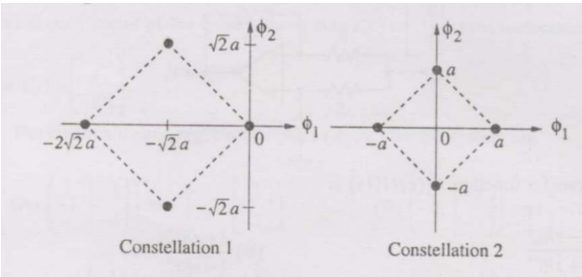

Fig. 3.0.10.1

a) The ratio of the average energy of Constellation 1 to that of Constellation 2 is:

i) $4a^2$ ii) 4 iii) 2 iv) 8

b) If these constellations are used over an AWGN channel, then:

i) Probability of symbol error for Constellation 1 is lower
ii) Probability of symbol error for Constellation 1 is higher
iii) Probability of symbol error is equal for both
iv) Value of N_0 will determine which has lower error

3.0.11 An input to a 6-level quantizer has PDF $f(x)$ as shown: it is piecewise-constant with three decision boundaries at -1, 0, and 1, chosen to maximize output entropy. The central plateau height is a, the outer flat region height is b. (EC 2007)

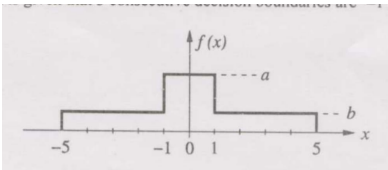

Fig. 3.0.11.1

a) The values of a and b are:

i) $a = \frac{1}{6},\ b = \frac{1}{12}$ ii) $a = \frac{1}{5},\ b = \frac{3}{40}$ iii) $a = \frac{1}{4},\ b = \frac{1}{16}$ iv) $a = \frac{1}{3},\ b = \frac{1}{24}$

b) Assuming reconstruction levels are midpoints of decision intervals, the ratio of signal power to quantization noise power is:

i) $\dfrac{152}{9}$ ii) $\dfrac{64}{3}$ iii) $\dfrac{76}{3}$ iv) 28